

分析报告

在一块可拉伸的电阻传感器上按压输入摩尔斯电码——轻/短敲是点(·)、重/长敲是划(—)。数字万用表记录电阻随时间的变化,本报告用这条曲线自动识别所敲的字母,并逐一给出每个字母的波形与识别准确率。

97.7%

点划逐次识别率

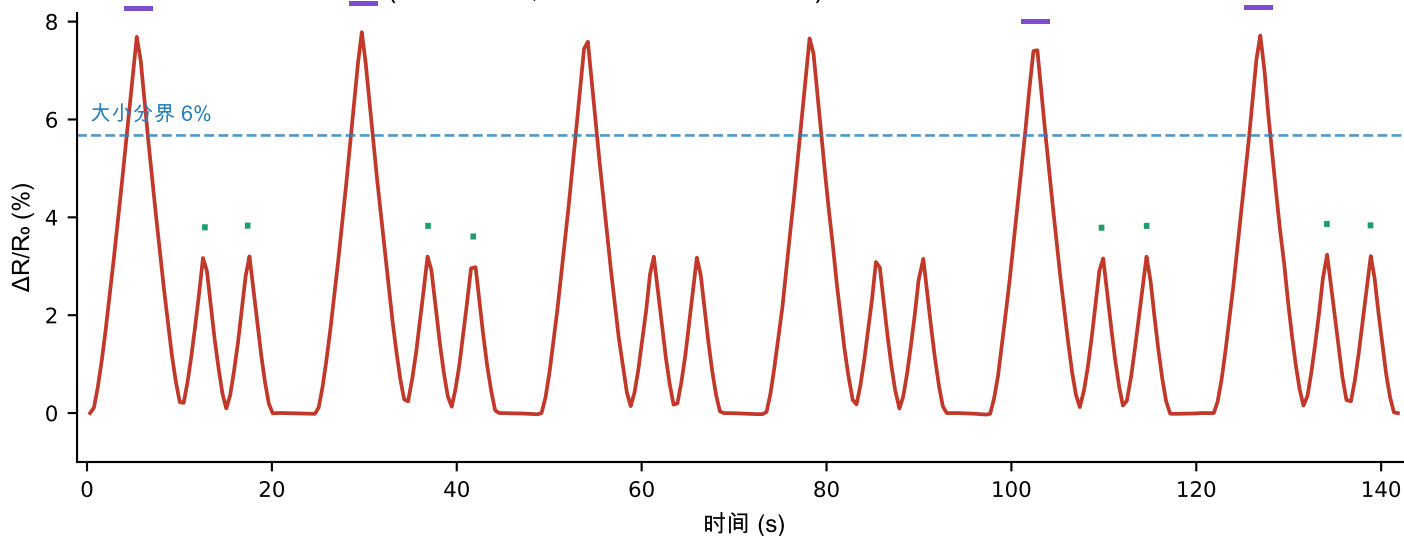
24 / 26

字母整体认对

1,672

参与统计的重复次数

示例:字母 D 的电阻波形(一大峰=划,两小峰=点 → —·· → D)



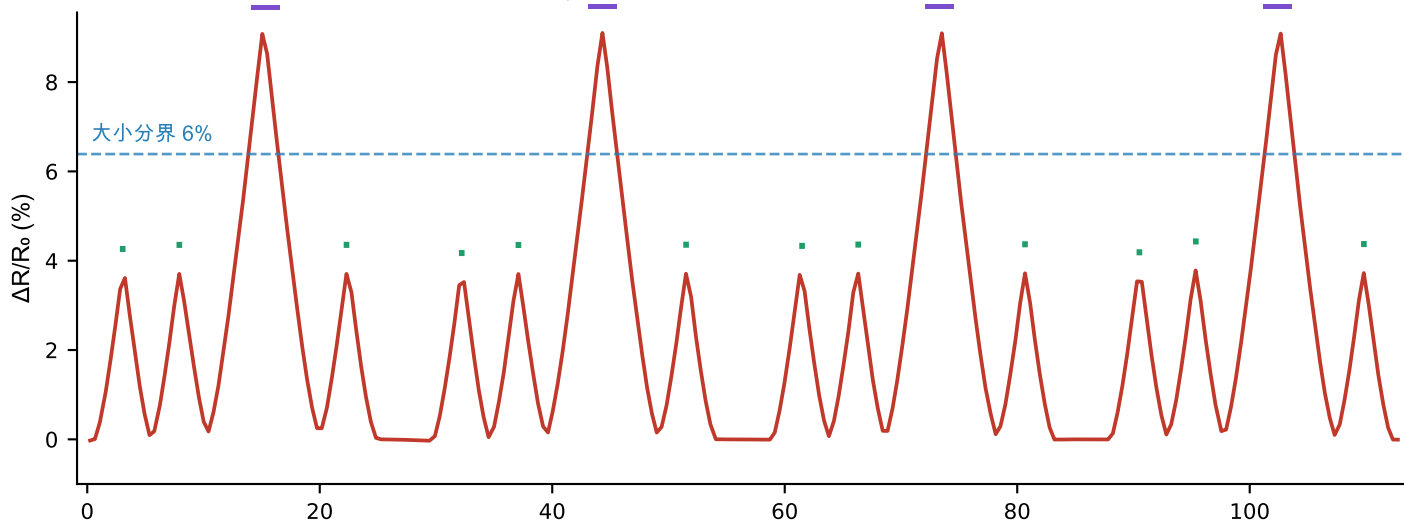
方法一句话:先减掉缓慢漂移的静息基线,找出每一次按压;一次按压有多「大」体现在峰更高、或按得更久。据此判点/划,再对照标准摩尔斯码表还原字母。

识别原理:怎么从波形认出点和划

一次敲击是否为「划」,看它比周围的敲击是不是更「大」。「大」有两条线索,程序按录制自身情况自动选用:

情况一·含点又含划的字母:比峰高

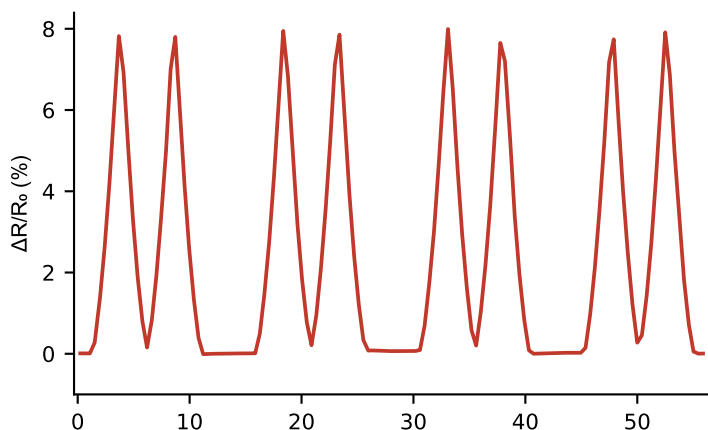
字母 F = ··· — 同一录制里划就是比点高,直接相对比较



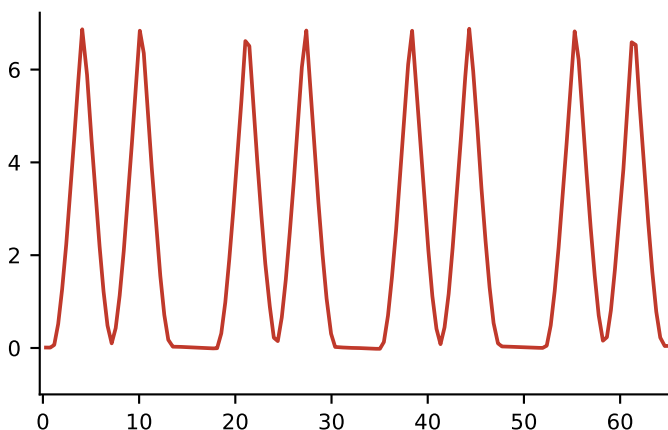
若对所有录制用同一个固定高度分界,只有 58% 正确(传感器软硬和按力不同);改成在同一录制内部相对比较,可达 95% 以上。

情况二·全是点或全是划的字母:比时长

I = ··· 时长≈3.4s



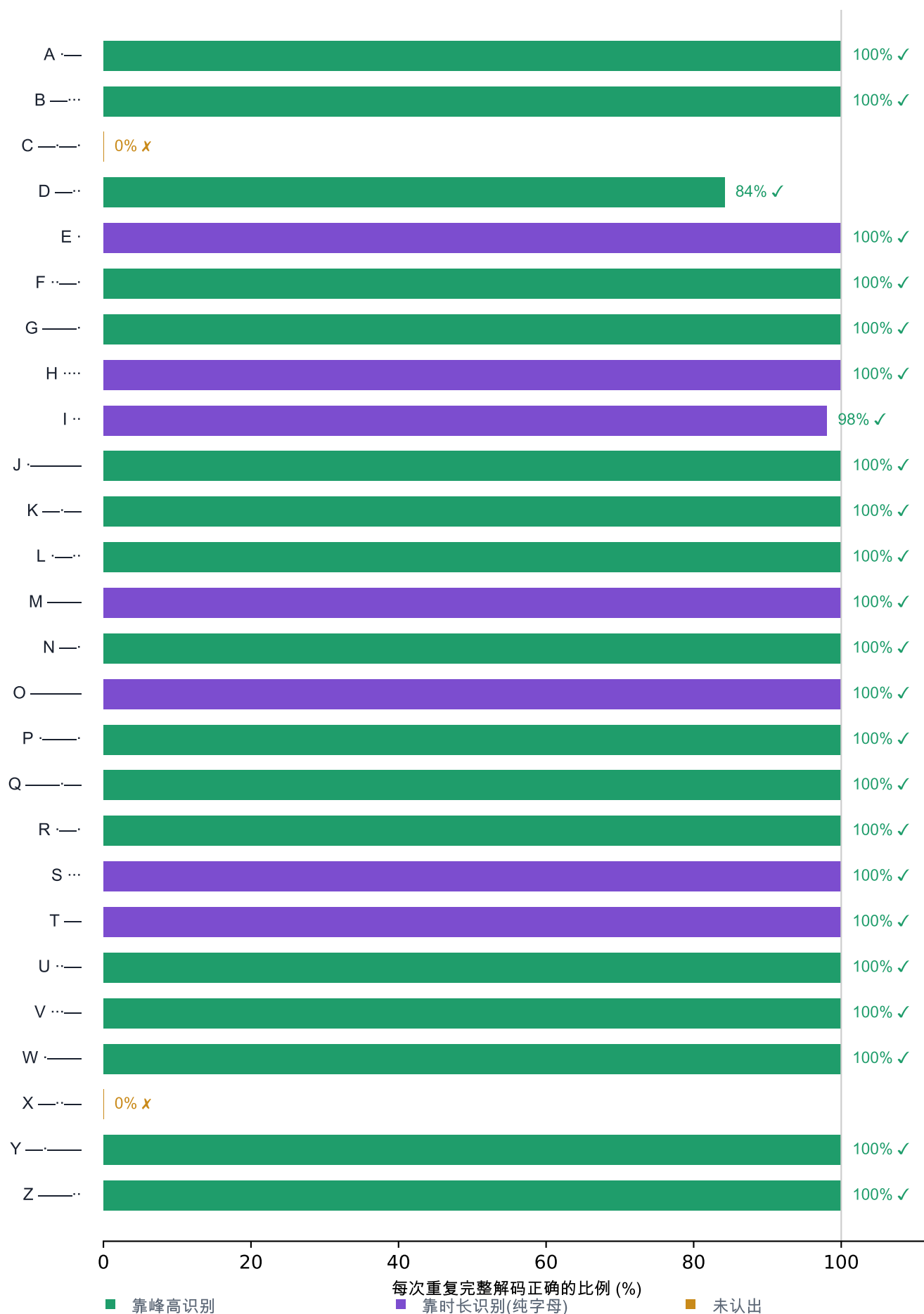
M = — 时长≈4.2s



纯字母一份录制里没有高矮对比。但划按得更久(点≈3秒、划≈4秒),而按压时长是人的动作、跨录制稳定。注意上图:点字母 I 的峰甚至比划字母 M 还高——所以此时不能靠高度,只能靠时长。

四个步骤:①找按压 → ②量峰高与时长 → ③小=点/大=划 → ④查摩尔斯码表。

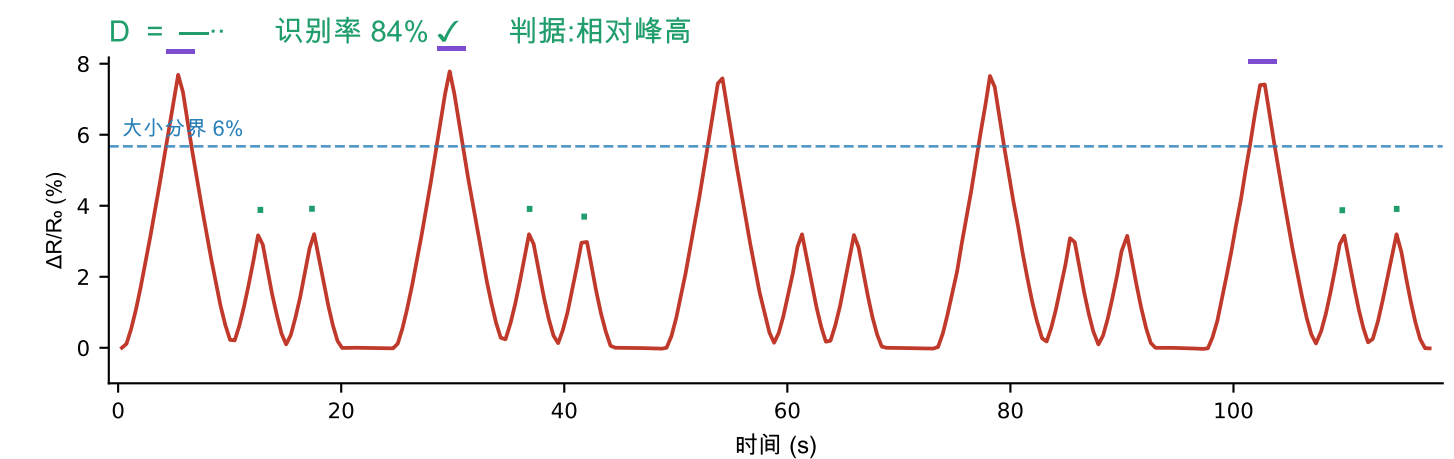
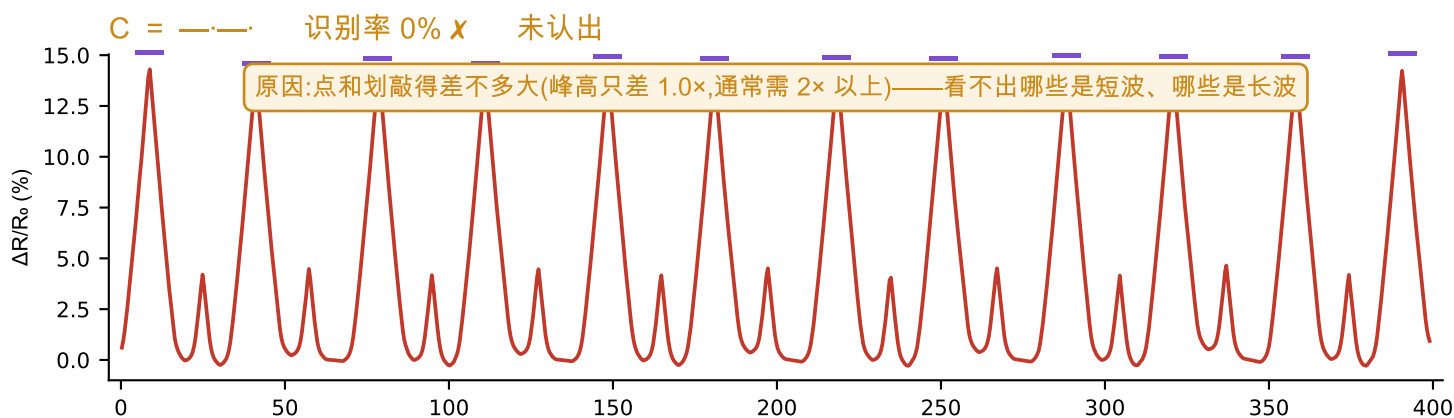
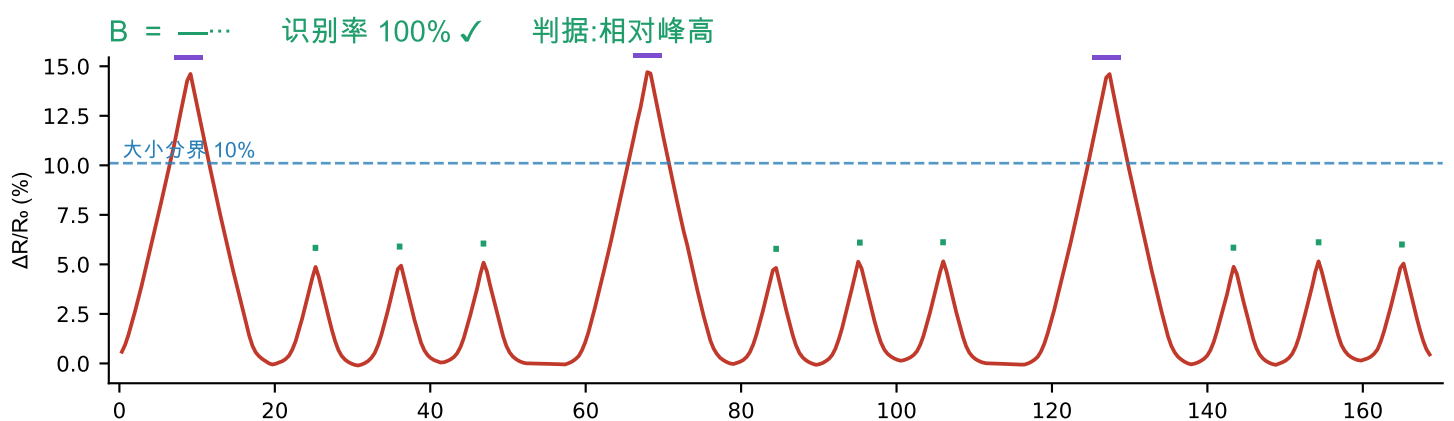
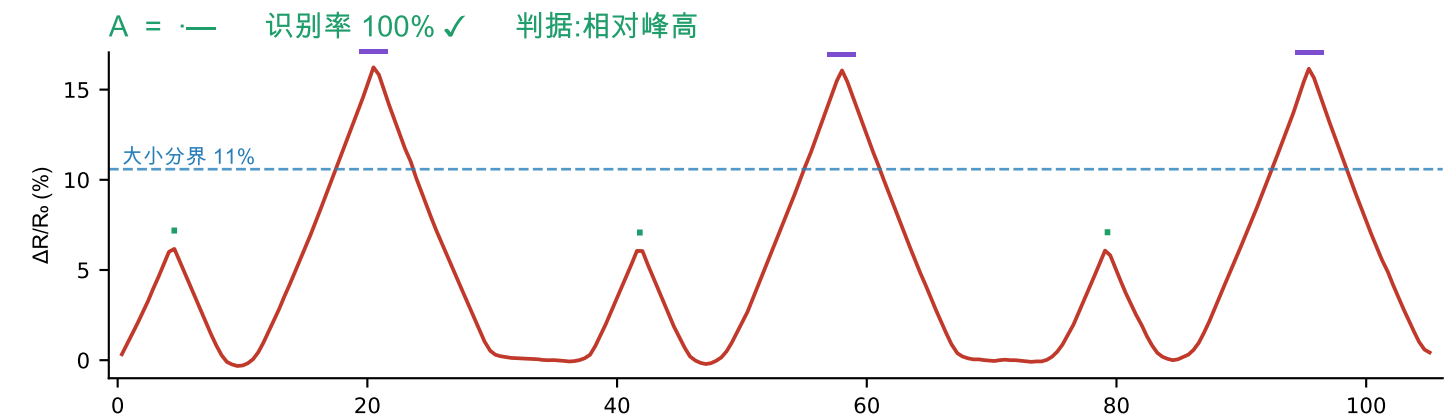
识别准确率:逐字母



未认出的 C, X 是因为那几份录制本身点划没做出区分(点按太重或划按太短),重录即可,不是方法问题。

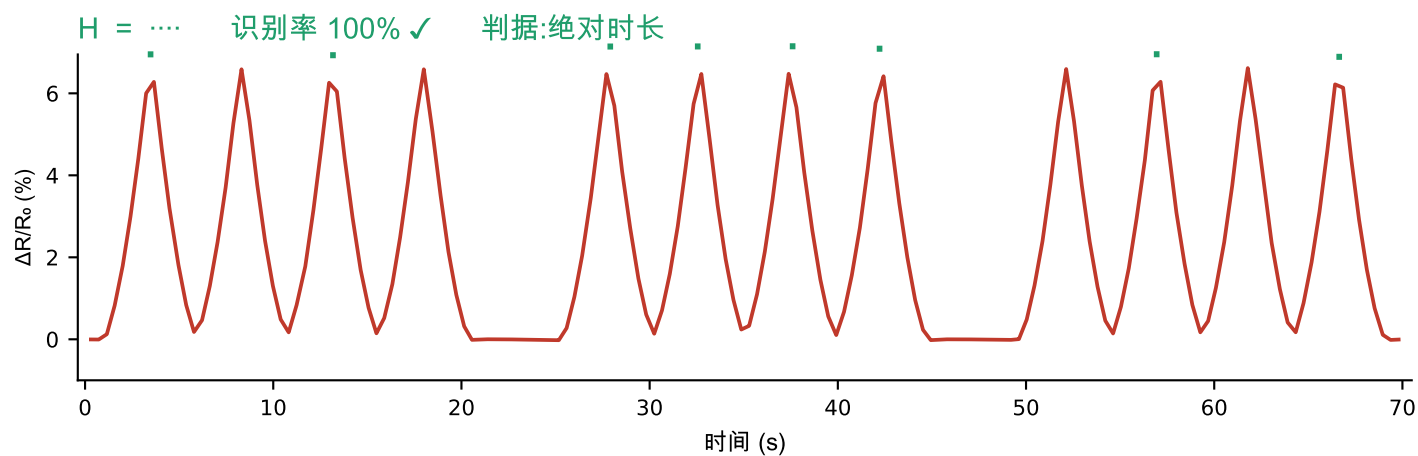
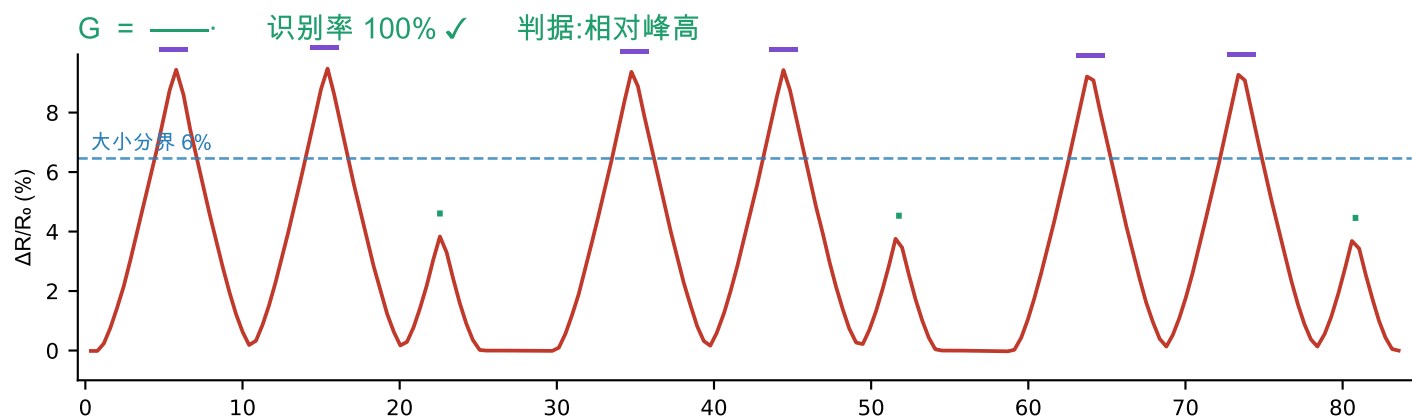
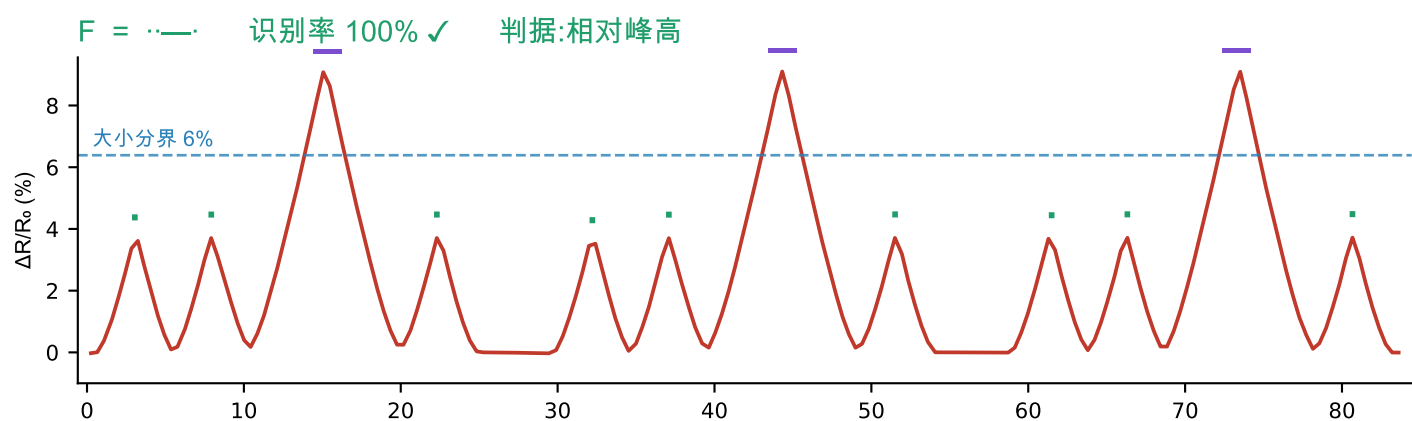
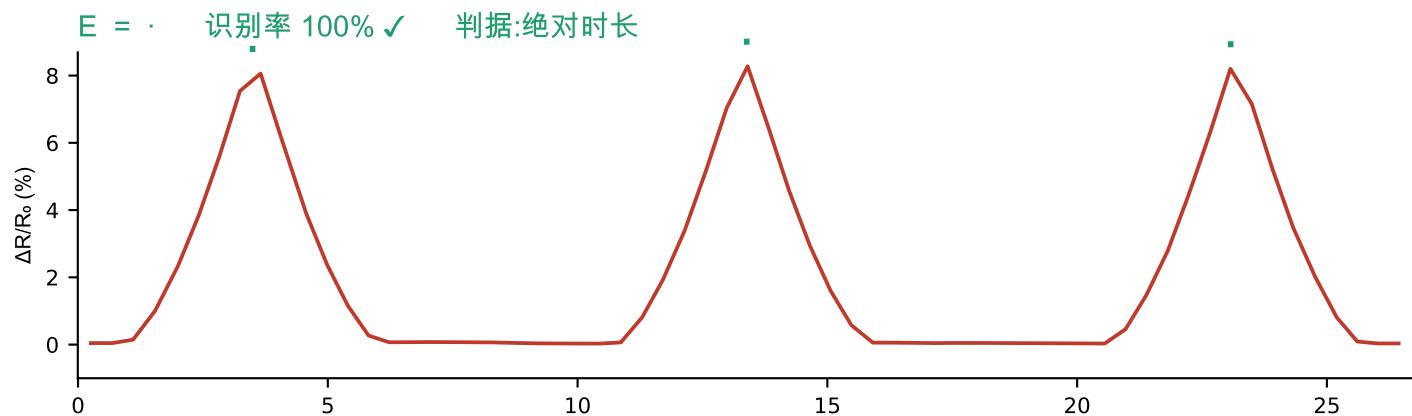
每个字母的波形

红=电阻曲线;绿点·为点、紫—为划,标在每次按压的峰顶。



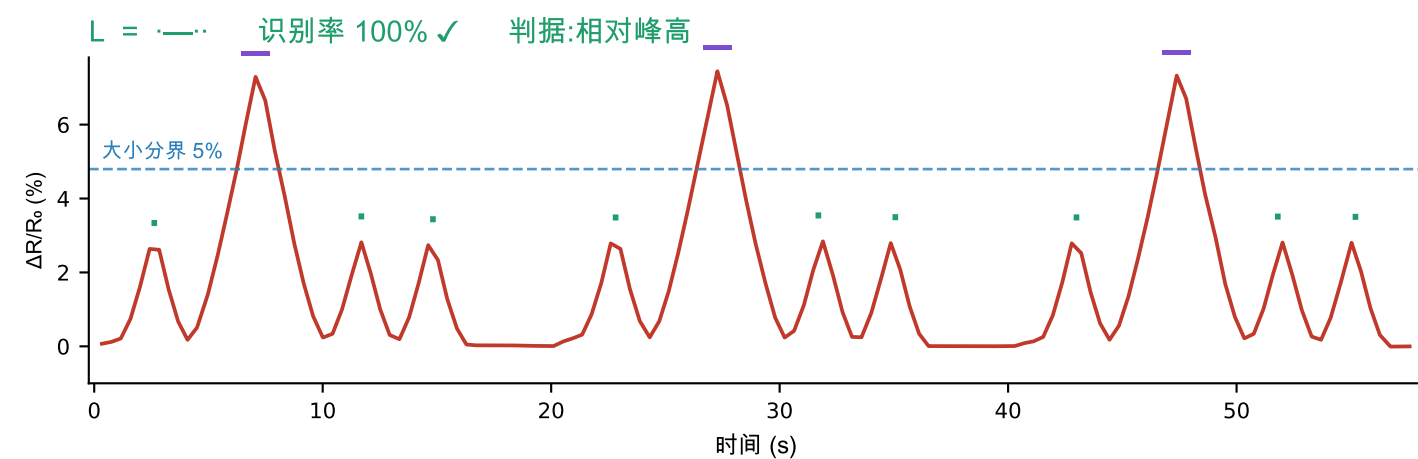
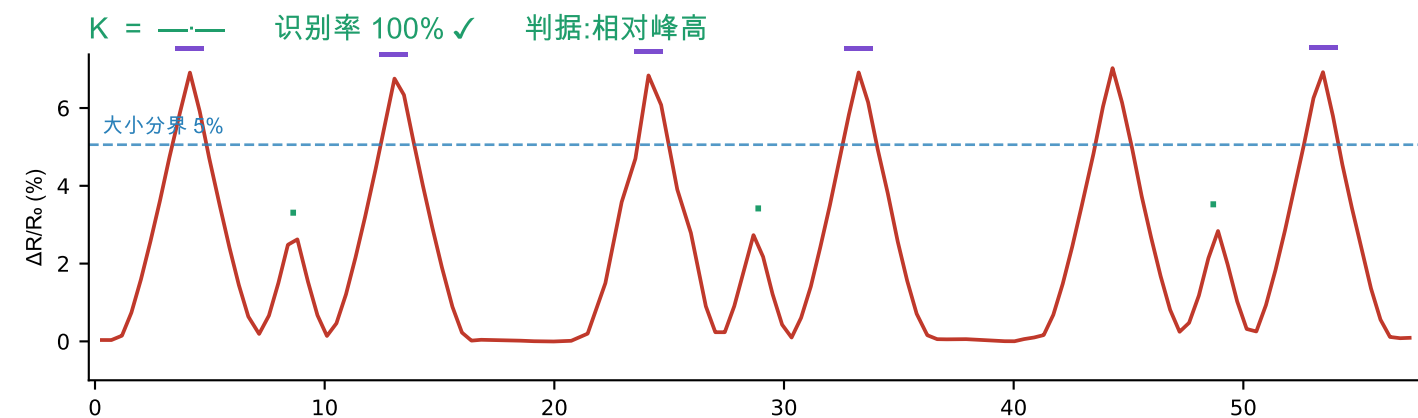
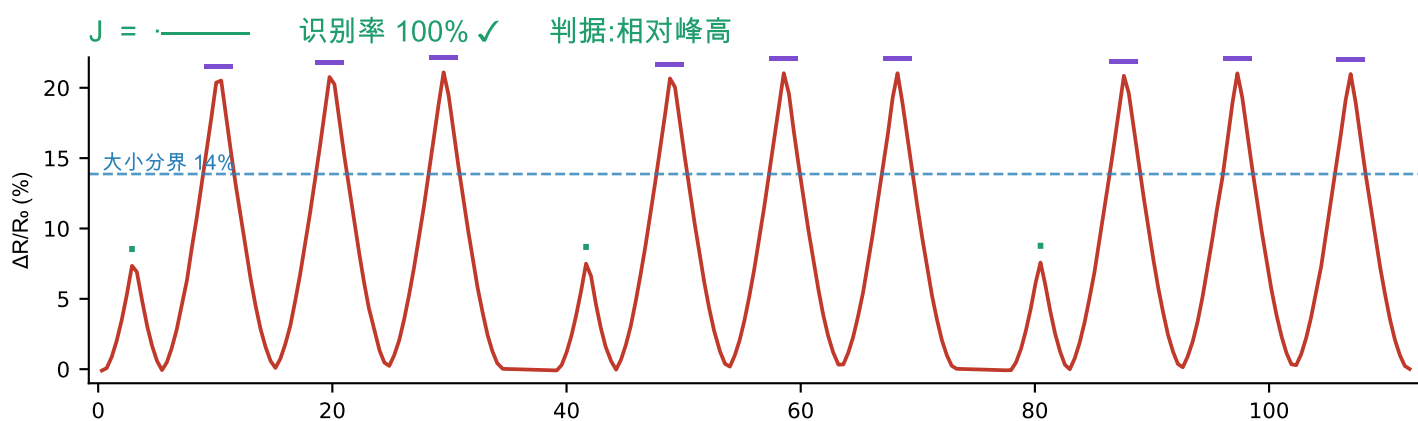
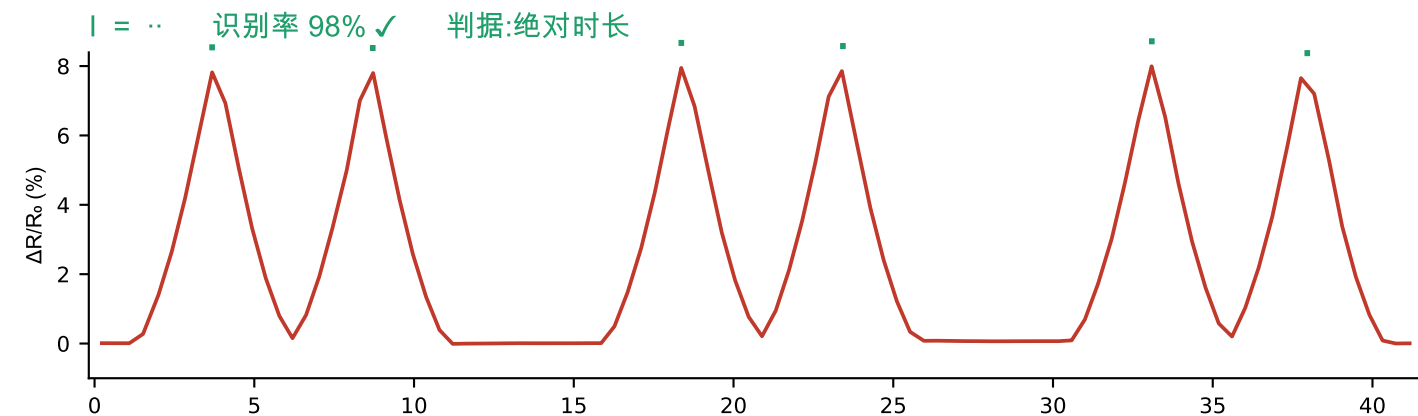
每个字母的波形

红=电阻曲线;绿点·为点、紫—为划,标在每次按压的峰顶。



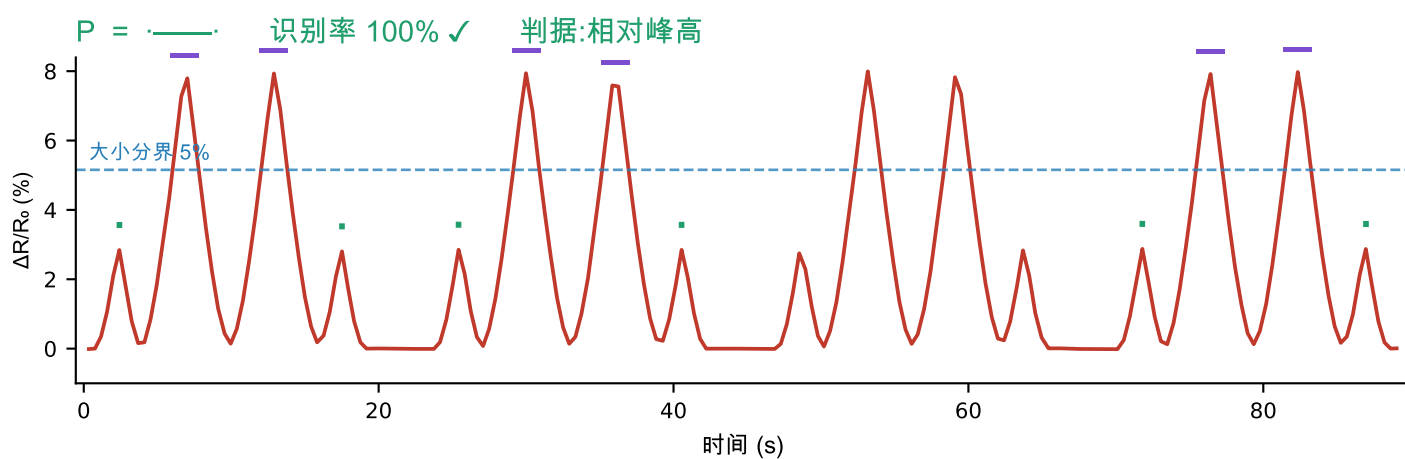
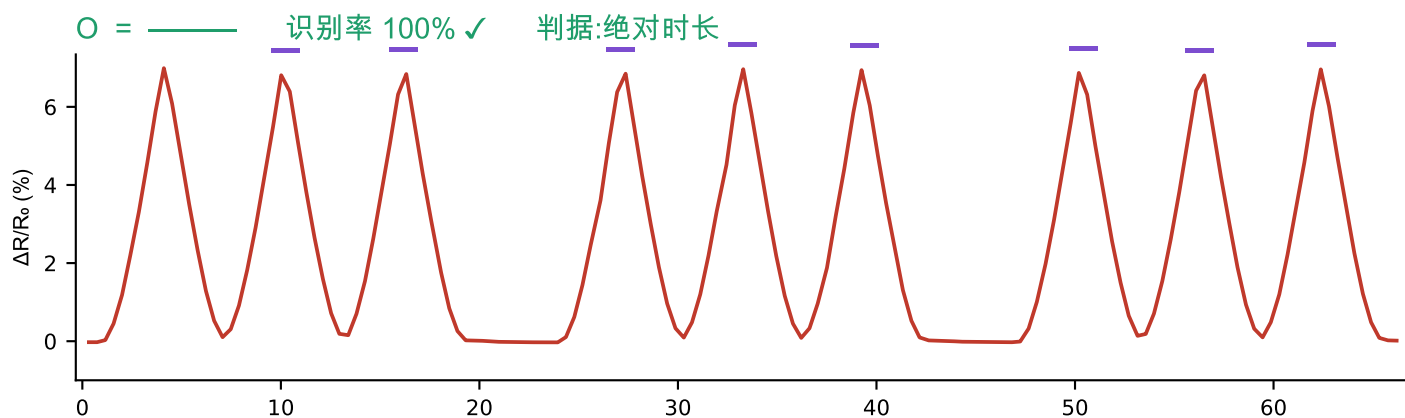
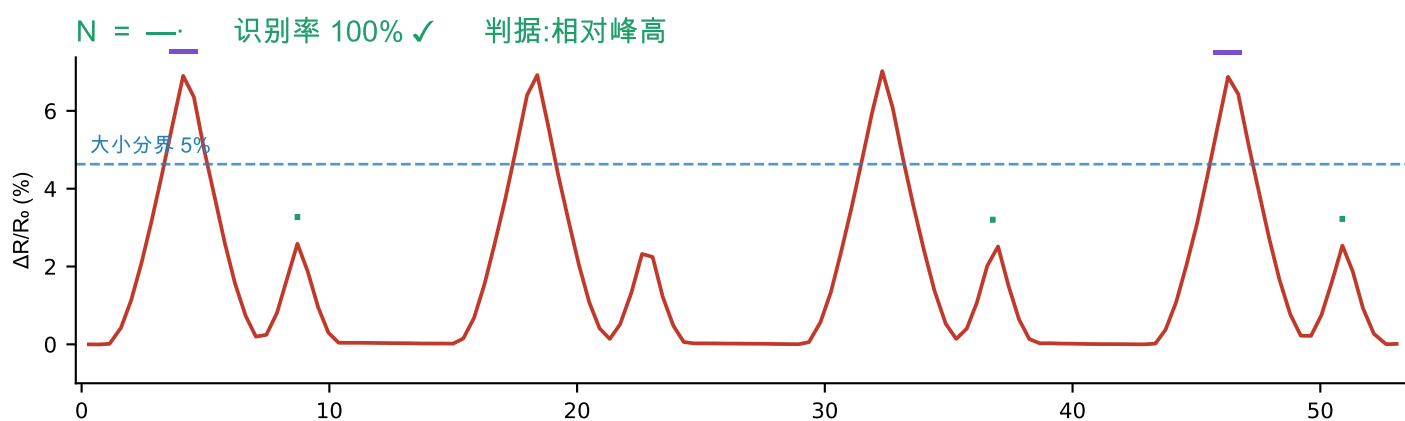
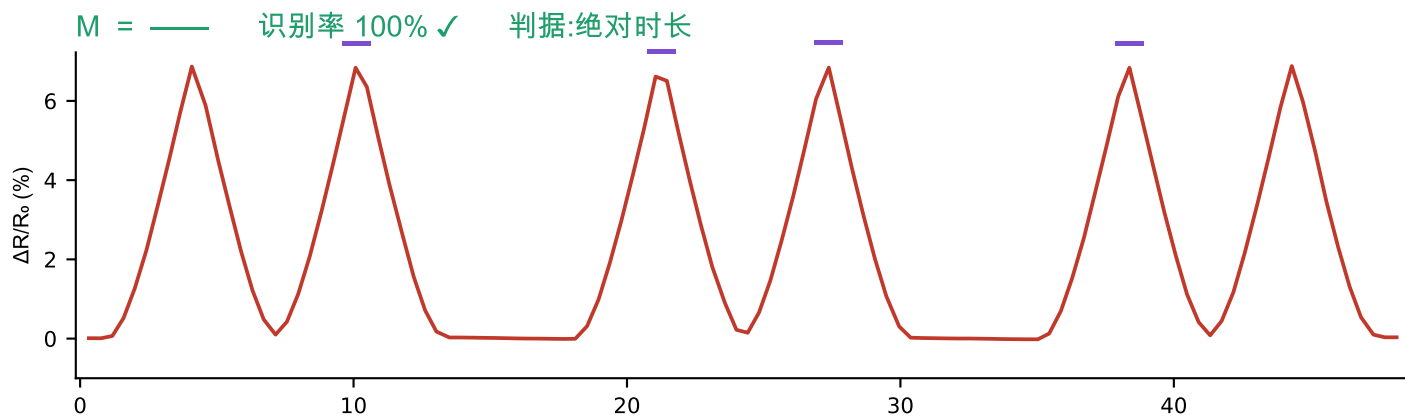
每个字母的波形

红=电阻曲线;绿点·为点、紫—为划,标在每次按压的峰顶。



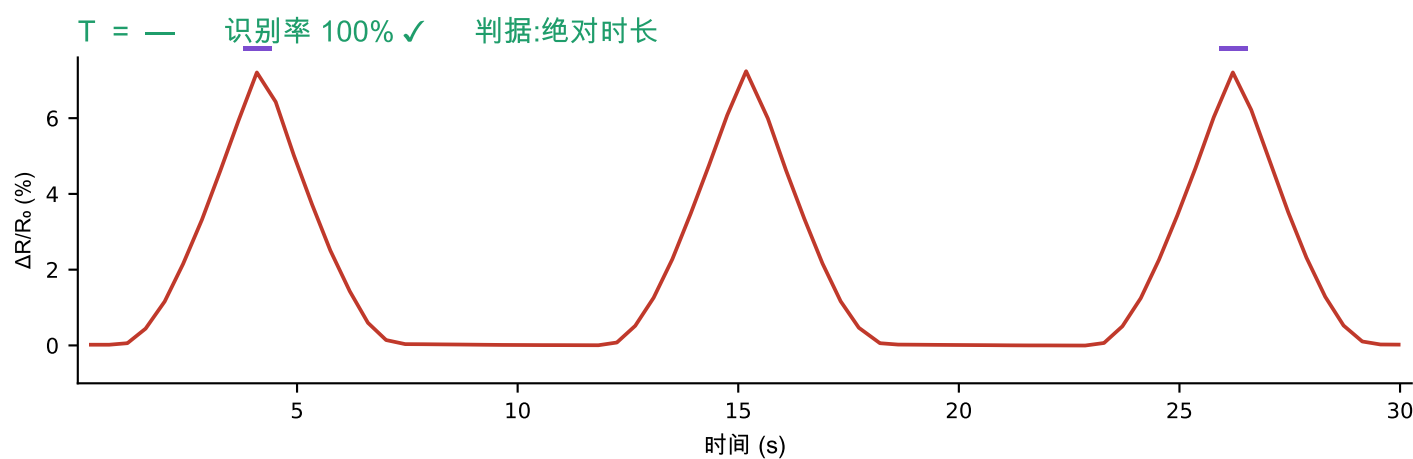
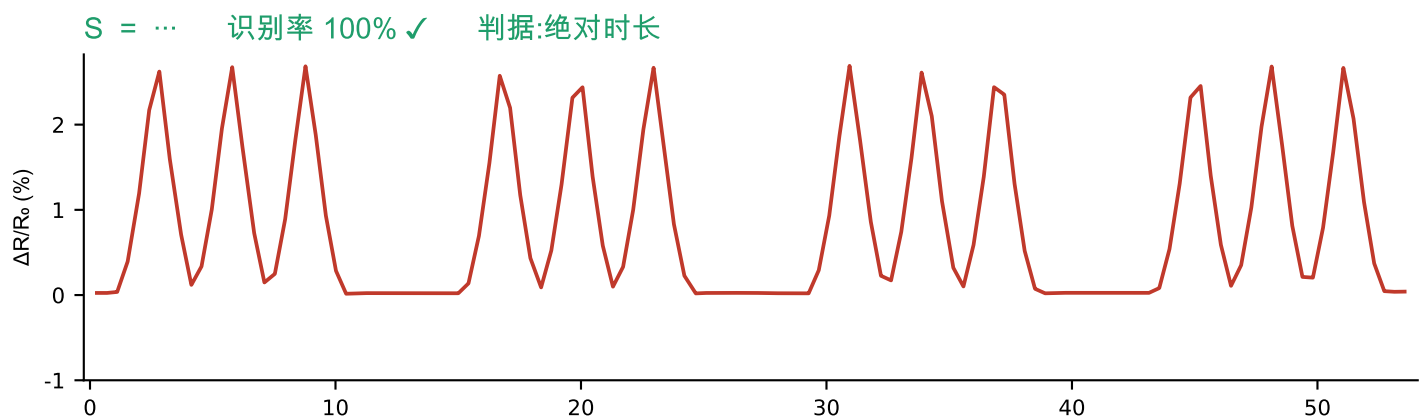
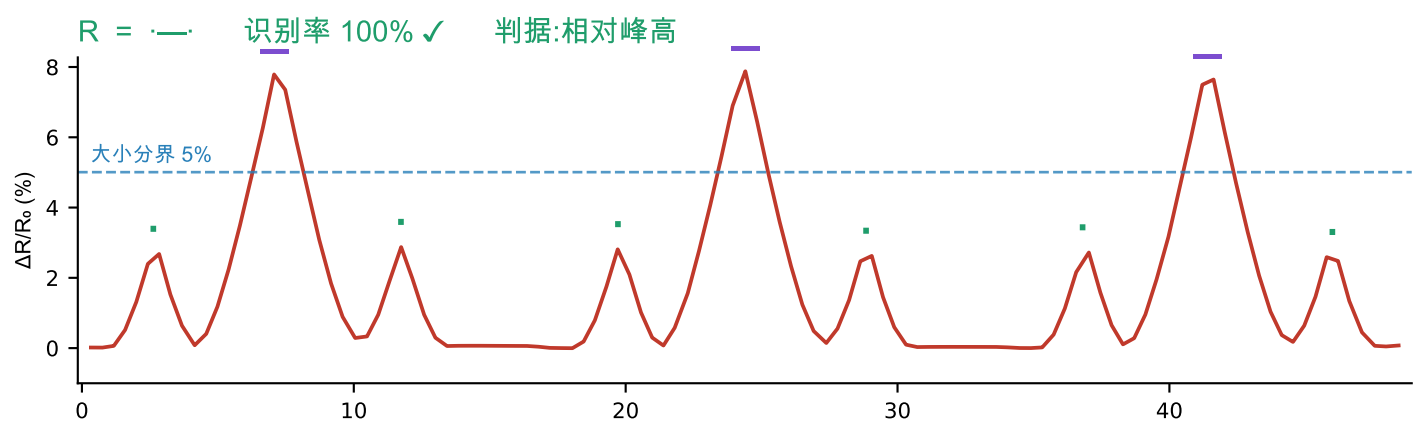
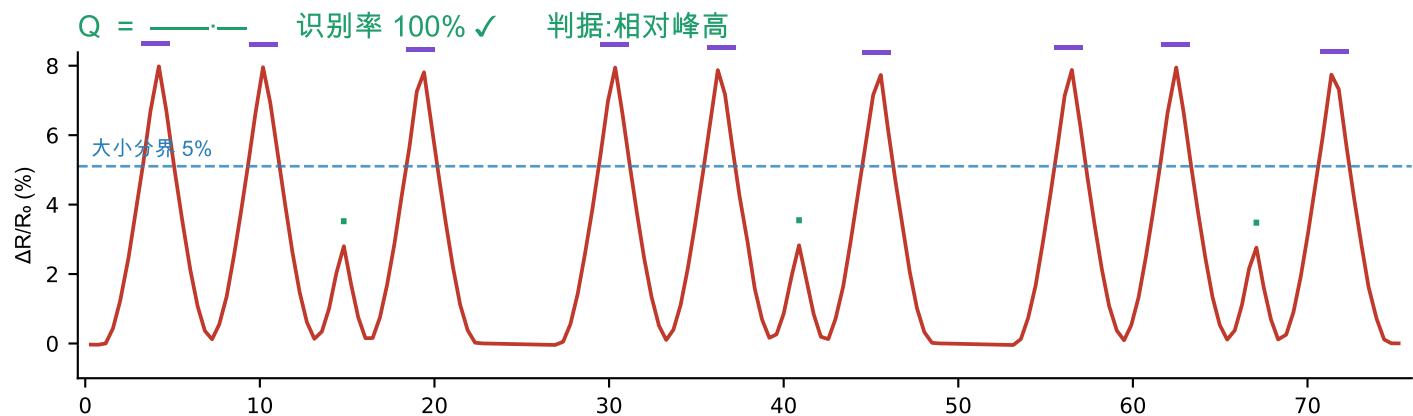
每个字母的波形

红=电阻曲线;绿点·为点、紫—为划,标在每次按压的峰顶。



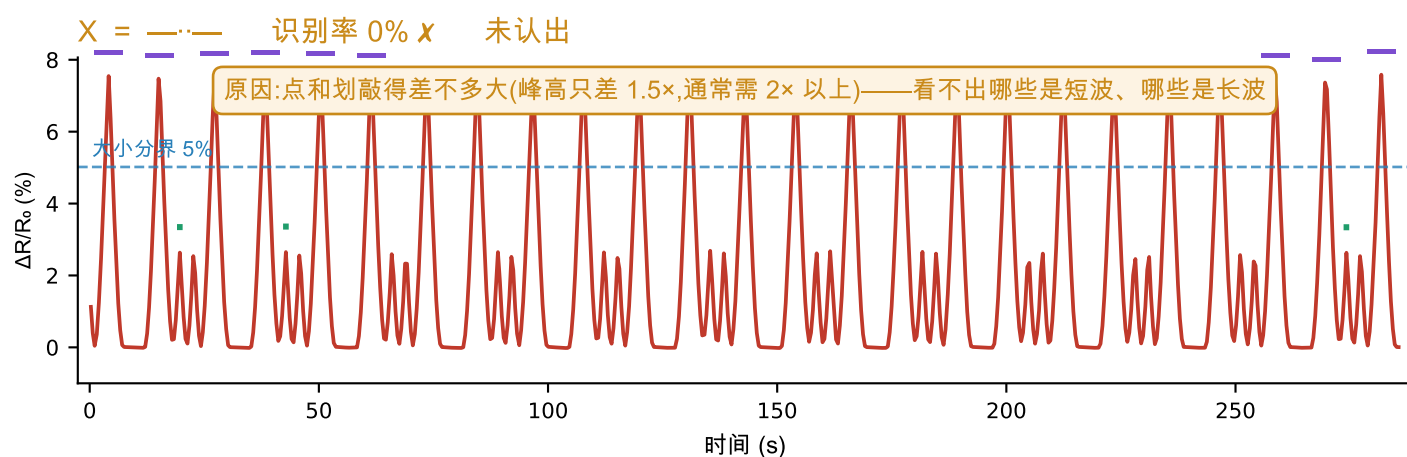
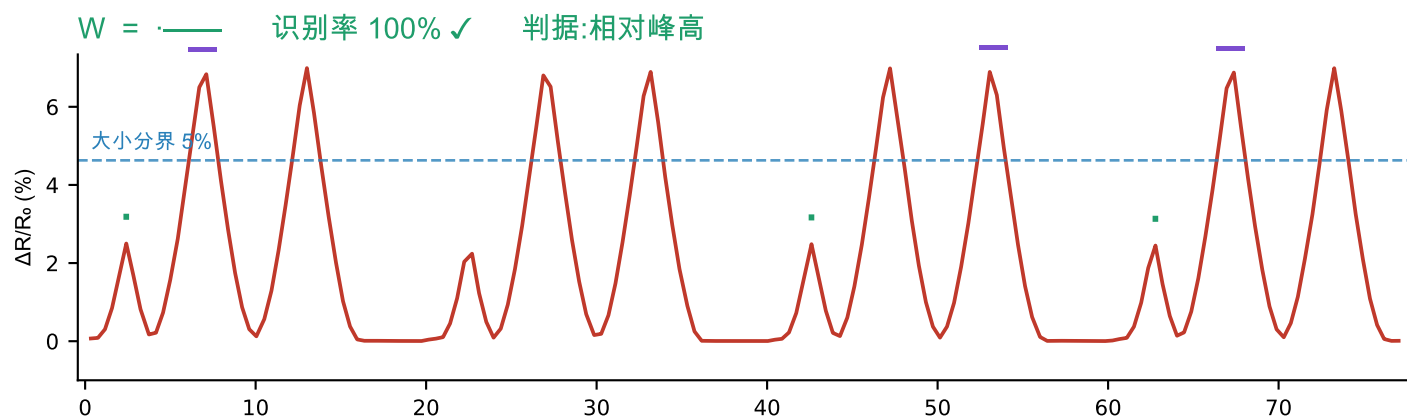
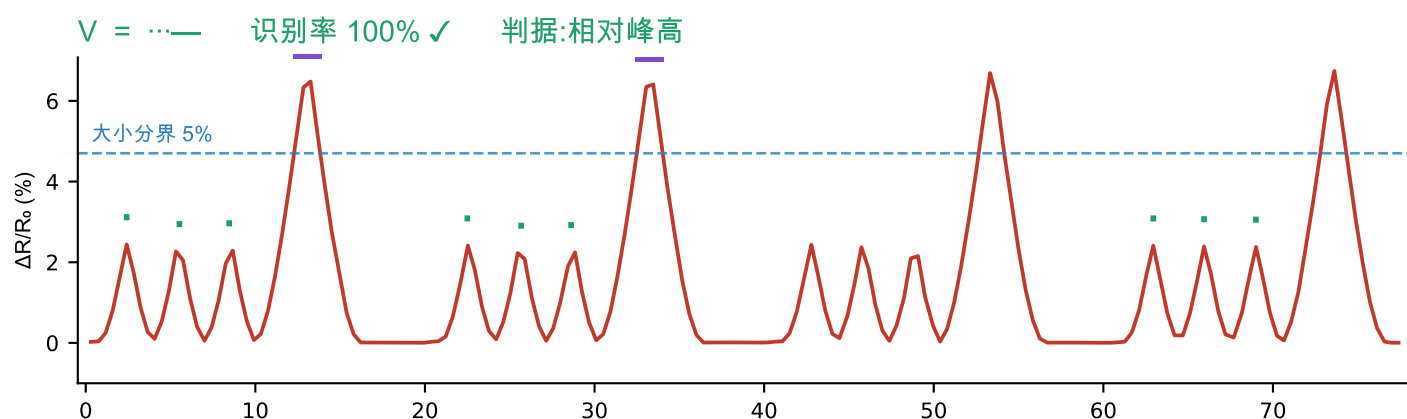
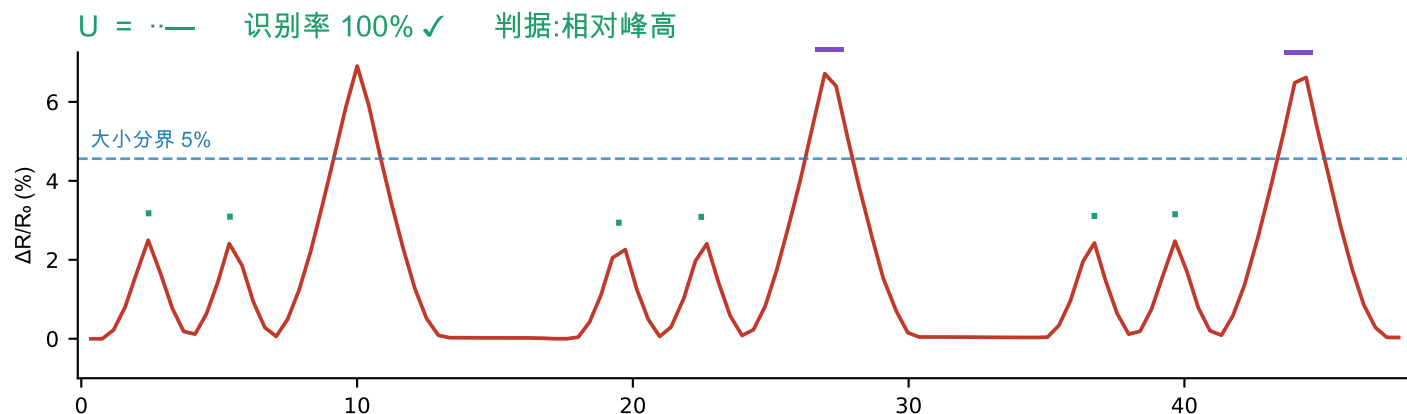
每个字母的波形

红=电阻曲线;绿点·为点、紫—为划,标在每次按压的峰顶。



每个字母的波形

红=电阻曲线;绿点·为点、紫—为划,标在每次按压的峰顶。



每个字母的波形

红=电阻曲线;绿点·为点、紫 — 为划,标在每次按压的峰顶。

